

$$7(a) \quad \lambda = v/f$$

$$\lambda_1 = \frac{340}{20000} = 0.017 \text{ m}$$

$$\lambda_2 = \frac{340}{80000} = 0.0043 \text{ m}$$

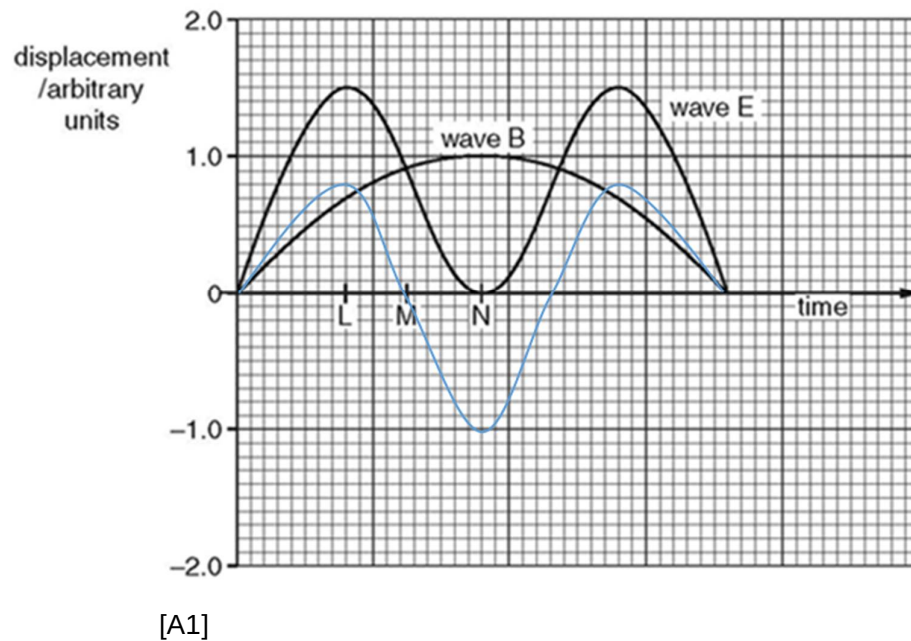
Range: 0.0043 to 0.017 m.

$$(b)(i) \quad x_L = 1.5 - 0.70 = 0.80$$

$$x_M = 0.90 - 0.90 = 0.00$$

$$x_N = 0.0 - 1.0 = -1.00$$

(ii)



(c)(i)
$$\frac{51.25 - 50.8}{50.8} = \frac{2v}{340}$$

$$v = 1.51 \text{ m s}^{-1}$$

(ii) Diffraction

(d)(i) The graph is linear with a negative gradient. Its y intercept = $2.8 \approx \ln 16$.

So the graph supports the equation $\ln I = -\alpha x + \ln 16$

$$I = 16e^{-\alpha x}$$

$$= I_0 e^{-\alpha x}$$

(ii) 1. m^{-1}

2.
$$\text{gradient} = \frac{2.80 - 0.85}{0.0 - 3.0} = -0.65 \text{ m}^{-1}$$

$$\therefore \alpha \approx 0.7 \text{ m}^{-1}$$

(iii) Draw a graph with a constant more negative (steeper) gradient.

The y-intercept should be the same as before.